

Cographs and Minimum Diamond-Generating Edge Sets in Boolean Lattices

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Abstract

We study a local closure operation on the cover edges of a Boolean lattice: whenever the two lower edges or the two upper edges of a square face are present, all four edges of that square are added. We prove that every set of cover edges generating the full cover graph of $\mathcal{B}_n$ has cardinality at least n , and we classify all generators attaining this bound. For a graph G on $[n]$, let

$$S_G = \{N_G(i) \rightarrow N_G(i) \cup \{i\} : i \in [n]\}.$$

Then S_G diamond-generates the full cover graph if and only if G is a cograph, and every minimum-cardinality generator arises uniquely in this way. Consequently, labeled minimum diamond-generating sets of $\mathcal{B}_n$ are in bijection with labeled cographs on n vertices.

Keywords. Boolean lattice; diamond closure; cograph; cover graph; graph twins; generating set.

1 Introduction

A diamond in the cover graph of a lattice is a square formed by two successive cover relations in either order. If two adjacent edges on one side of the square are known, a natural completion rule is to add the two edges on the other side. This diamond operation appears in the study of rational relations among pseudo-roots of noncommutative polynomials; in particular, Retakh and Saks study diamond closure in cover graphs of modular lattices [2]. The present paper isolates a finite extremal question for the Boolean lattice.

Let $\mathcal{B}_n$ denote the Boolean lattice of subsets of $[n]$. We ask: what is the smallest number of cover edges from which every cover edge of $\mathcal{B}_n$ can be obtained by repeated diamond completions, and what do all smallest generating sets look like? The answer is governed by cographs, the class of graphs generated from single vertices by disjoint union and join [1]. Cographs are also characterized as the finite graphs with no induced four-vertex path [3].

For a graph G on a finite vertex set X , write $N_G(i)$ for the open neighborhood of i , and associate to G the set of cover edges

$$S_G = \{N_G(i) \rightarrow N_G(i) \cup \{i\} : i \in X\}.$$

Our main result is the following.

Theorem 1.1 (Main classification). *Let X be a finite nonempty set.*

- (i) *Every diamond-generating set of cover edges of $\mathcal{B}_X$ has cardinality at least $|X|$, and this bound is attained.*

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- (ii) For a graph G on X , the set S_G diamond-generates $\mathcal{B}_X$ if and only if G is a cograph.
- (iii) Every diamond-generating set of cardinality $|X|$ is equal to S_G for a unique cograph G on X .

The lower bound follows because a diamond completion cannot introduce a coordinate direction that was absent from the initial set. A generator of minimum cardinality therefore contains exactly one edge in each direction. A coordinate-projection lemma then forces a pairwise symmetry among the lower endpoints of those edges, so they are precisely the open neighborhoods of a simple graph. Cographs generate by induction through their disjoint-union and join constructions. Conversely, generation passes to every induced subgraph, while any seed set that begins to expand must contain two edges forming one half of a diamond; for neighborhood seeds, such a pair is exactly a pair of false or true twins. The hereditary twin characterization of cographs then completes the proof.

Throughout, graphs are finite, simple, and undirected. The word *minimum* refers to minimum cardinality, rather than inclusion-minimality.

2 Diamond closure and coordinate projection

For a finite set X , let $\mathcal{B}_X = 2^X$ be the Boolean lattice ordered by inclusion. When $X = [n] = \{1, \dots, n\}$, we also write $\mathcal{B}_n$.

A cover edge of $\mathcal{B}_X$ has the form

$$e_i(A) : A \longrightarrow A \cup \{i\}, \quad A \subseteq X \setminus \{i\}.$$

The element i is called the *direction* of the edge. We write $E(\mathcal{B}_X)$ for the full set of cover edges.

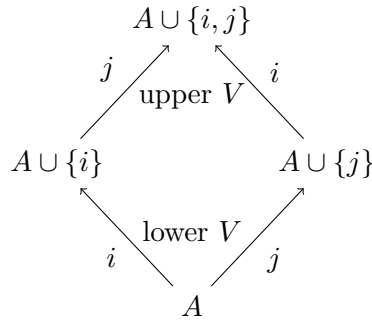
A diamond is determined by a set $A \subseteq X$ and distinct elements $i, j \in X \setminus A$. Its lower and upper halves are, respectively,

$$L(A; i, j) = \{e_i(A), e_j(A)\}$$

and

$$U(A; i, j) = \{e_i(A \cup \{j\}), e_j(A \cup \{i\})\}.$$

Thus its four-edge set is $L(A; i, j) \cup U(A; i, j)$.



Definition 2.1 (Synchronous diamond closure). Let $S \subseteq E(\mathcal{B}_X)$. Define an increasing sequence $S^{(0)}, S^{(1)}, \dots$ by $S^{(0)} = S$ and by letting $S^{(t+1)}$ be $S^{(t)}$ together with all four edges of every diamond for which either $L(A; i, j) \subseteq S^{(t)}$ or $U(A; i, j) \subseteq S^{(t)}$. The *diamond closure* of S is

$$\text{cl}_{\diamond, X}(S) = \bigcup_{t \geq 0} S^{(t)}.$$

We call S *diamond-generating* if $\text{cl}_{\diamond, X}(S) = E(\mathcal{B}_X)$.

Because $E(\mathcal{B}_X)$ is finite, the sequence in Definition 2.1 stabilizes. Its stable value is equivalently the smallest set of cover edges containing S that is closed under diamond completion.

We next record the projection property that will be used twice in the proof. For $Y \subseteq X$, define the coordinate projection of an edge in a retained direction by

$$\pi_Y(e_i(A)) = e_i(A \cap Y), \quad i \in Y,$$

and discard edges whose direction is not in Y . For a set T of edges, let $\pi_Y(T)$ be the set of all retained projected edges.

Lemma 2.2 (Coordinate projection). *For every $S \subseteq E(\mathcal{B}_X)$ and every $Y \subseteq X$,*

$$\pi_Y(\text{cl}_{\diamond, X}(S)) \subseteq \text{cl}_{\diamond, Y}(\pi_Y(S)).$$

Consequently, if S diamond-generates $\mathcal{B}_X$, then $\pi_Y(S)$ diamond-generates $\mathcal{B}_Y$.

Proof. Let $S^{(t)}$ be the closure sequence in $\mathcal{B}_X$, and let $T^{(t)}$ be the closure sequence in $\mathcal{B}_Y$ starting from $T^{(0)} = \pi_Y(S)$. We prove by induction on t that

$$\pi_Y(S^{(t)}) \subseteq T^{(t)}.$$

The assertion is immediate for $t = 0$.

Consider a diamond used in passing from $S^{(t)}$ to $S^{(t+1)}$, with directions i and j . If $i, j \in Y$, its four edges project to the four edges of a diamond in $\mathcal{B}_Y$, and the triggering lower or upper V projects to the corresponding triggering V . Hence every projected edge added by this completion belongs to $T^{(t+1)}$.

If exactly one of i, j lies in Y , the two edges in the retained direction project to the same edge of $\mathcal{B}_Y$. Completing the diamond therefore produces no new projected edge: the projection of the newly added retained-direction edge is already the projection of the retained-direction edge in the triggering V . If neither direction lies in Y , all four edges are discarded. These cases establish the induction step.

Taking the union over t gives the stated inclusion. If S generates $\mathcal{B}_X$, then

$$E(\mathcal{B}_Y) = \pi_Y(E(\mathcal{B}_X)) = \pi_Y(\text{cl}_{\diamond, X}(S)) \subseteq \text{cl}_{\diamond, Y}(\pi_Y(S)),$$

and the reverse inclusion in $E(\mathcal{B}_Y)$ is automatic. □

3 Minimum size and graph encoding

Lemma 3.1 (Direction support). *If no edge of $S \subseteq E(\mathcal{B}_X)$ has direction i , then no edge of $\text{cl}_{\diamond, X}(S)$ has direction i .*

Proof. Every diamond has two directions, and each of its lower and upper halves contains one edge in each direction. Thus a diamond completion can produce a new edge in direction i only when its triggering half already contains an edge in direction i . Induction on the closure rounds proves the claim. □

Proposition 3.2 (Minimum size). *Every diamond-generating subset of $E(\mathcal{B}_X)$ has cardinality at least $|X|$, and this bound is attained.*

Proof. By Lemma 3.1, a generating set must contain at least one edge in each of the $|X|$ directions. For sharpness, consider the bottom star

$$S_{\text{bot}} = \{e_i(\emptyset) : i \in X\}.$$

We prove by induction on r that every edge $e_i(A)$ with $|A| \leq r$ lies in $\text{cl}_{\diamond, X}(S_{\text{bot}})$. The case $r = 0$ is the definition of S_{bot} . Suppose the assertion holds through r , and let $|A| = r + 1$. Choose $j \in A$ and set $B = A \setminus \{j\}$. Since $i \notin A$, the two edges $e_i(B)$ and $e_j(B)$ are available by the induction hypothesis. They form the lower half of a diamond, whose upper half contains

$$e_i(B \cup \{j\}) = e_i(A).$$

Thus all cover edges are generated.

Dually, the top star

$$S_{\text{top}} = \{e_i(X \setminus \{i\}) : i \in X\}$$

also generates. Hence the minimum cardinality is $|X|$. $\square$

Let S now be a diamond-generating set of cardinality $|X|$. It contains exactly one edge in each direction, so there are uniquely determined sets $A_i \subseteq X \setminus \{i\}$ such that

$$S = \{e_i(A_i) : i \in X\}.$$

Lemma 3.3 (Symmetry lemma). *For distinct $i, j \in X$,*

$$j \in A_i \iff i \in A_j.$$

Proof. Project to the two-coordinate Boolean lattice on $Y = \{i, j\}$. By Lemma 2.2, $\pi_Y(S)$ generates $\mathcal{B}_Y$. All seed edges in directions outside Y are discarded, so $\pi_Y(S)$ consists of exactly one edge in direction i and one edge in direction j . More precisely,

$$\pi_Y(e_i(A_i)) = \begin{cases} e_i(\emptyset), & j \notin A_i, \\ e_i(\{j\}), & j \in A_i, \end{cases}$$

and analogously for the direction- j edge.

The lattice $\mathcal{B}_Y$ has a single diamond. A pair containing one edge in each direction generates that diamond exactly when the two edges form its lower half or its upper half. If one is lower and the other is upper, no diamond completion is available and the closure does not expand. Therefore the two projected edges must both be lower or both be upper, which is precisely

$$j \in A_i \iff i \in A_j.$$

$\square$

Corollary 3.4 (Graph encoding). *Every diamond-generating set of cardinality $|X|$ is equal to S_G for a unique graph G on X .*

Proof. Define adjacency for distinct $i, j \in X$ by

$$ij \in E(G) \iff j \in A_i.$$

The relation is symmetric by Lemma 3.3, and it has no loops because $i \notin A_i$. Hence it defines a simple graph. Its open neighborhoods are $N_G(i) = A_i$, so $S = S_G$. The neighborhoods determine every adjacency, which gives uniqueness. $\square$

Example 3.5. For the empty graph on X , every neighborhood is empty and $S_G = S_{\text{bot}}$. For the complete graph on X , every neighborhood is $X \setminus \{i\}$ and $S_G = S_{\text{top}}$.

4 Cographs and twins

The join $G_1 \vee G_2$ is obtained from the disjoint union of G_1 and G_2 by adding every edge between their vertex sets. A *cograph* is a graph obtained from one-vertex graphs by repeated disjoint union and join. This is equivalent to the standard definition as a graph with no induced copy of the four-vertex path P_4 [1, 3].

Two distinct vertices x, y are *false twins* if they are nonadjacent and have the same neighbors outside $\{x, y\}$. They are *true twins* if they are adjacent and have the same neighbors outside $\{x, y\}$. Either type will be called a pair of *twins*.

We use the following standard characterization; a proof is included for completeness.

Lemma 4.1 (Hereditary twin characterization). *A finite graph G is a cograph if and only if every induced subgraph of G with at least two vertices contains a pair of twins.*

Proof. Suppose first that G is a cograph. We prove the hereditary statement by induction on $|V(G)|$. It is immediate for a one-vertex graph. A nontrivial cograph has a construction whose final operation expresses it either as a disjoint union

$$G = G_1 \sqcup \cdots \sqcup G_k$$

or as a join

$$G = G_1 \vee \cdots \vee G_k,$$

where $k \geq 2$ and each G_a is a smaller cograph.

Let H be an induced subgraph of G with at least two vertices, and put $H_a = H[V(H) \cap V(G_a)]$. If some H_a has at least two vertices, then it has twins by induction; those vertices remain twins in H , because every vertex outside G_a is adjacent to both of them in the join case and to neither of them in the disjoint-union case. If every nonempty H_a is a singleton, choose vertices in two distinct parts. They are false twins in the disjoint-union case and true twins in the join case.

Conversely, suppose every induced subgraph of G with at least two vertices has twins. We induct on $|V(G)|$. Choose twins x, y in G and let $H = G - y$. The graph H has the same hereditary twin property and is therefore a cograph by induction. Take a construction tree for H . Replace the leaf corresponding to x by $K_1 \sqcup K_1$ if x, y are false twins, and by $K_1 \vee K_1$ if they are true twins. Leaving all subsequent union and join operations unchanged gives exactly G , so G is a cograph. $\square$

5 Cographs generate

Proposition 5.1. *If G is a cograph on X , then S_G diamond-generates $\mathcal{B}_X$.*

Proof. We induct on $|X|$. The assertion is immediate when $|X| = 1$. Suppose $|X| \geq 2$. By the recursive definition of a cograph, there is a partition $X = X_1 \sqcup \cdots \sqcup X_k$, with $k \geq 2$, such that each induced graph $G_a = G[X_a]$ is a smaller cograph and G is either their disjoint union or their join.

First suppose $G = G_1 \sqcup \cdots \sqcup G_k$. For each a , consider the Boolean face

$$F_a^0 = \{A \subseteq X : A \subseteq X_a\},$$

which is naturally isomorphic to $\mathcal{B}_{X_a}$. If $i \in X_a$, then $N_G(i) = N_{G_a}(i)$, so the seeds of S_G in directions belonging to X_a are precisely the embedded copy of S_{G_a} in this face. Every diamond of the face is also a diamond of $\mathcal{B}_X$. By induction, these seeds generate every cover edge in F_a^0 , and in particular they generate $e_i(\emptyset)$ for every $i \in X_a$. Applying this to all parts produces the full bottom star S_{bot} , which generates $\mathcal{B}_X$ by Proposition 3.2.

Now suppose $G = G_1 \vee \cdots \vee G_k$. For each a , consider the face

$$F_a^1 = \{(X \setminus X_a) \cup A : A \subseteq X_a\},$$

again naturally isomorphic to $\mathcal{B}_{X_a}$. For $i \in X_a$,

$$N_G(i) = (X \setminus X_a) \cup N_{G_a}(i),$$

so the corresponding seeds form the embedded copy of S_{G_a} in F_a^1 . Induction generates the top star of this face. Its direction- i edge is

$$e_i(X \setminus \{i\}) : X \setminus \{i\} \longrightarrow X.$$

Taking all parts yields the full top star S_{top} , which also generates $\mathcal{B}_X$. $\square$

6 The converse

Lemma 6.1 (Induced-subgraph inheritance). *Let G be a graph on X , and let $Y \subseteq X$. If S_G diamond-generates $\mathcal{B}_X$, then $S_{G[Y]}$ diamond-generates $\mathcal{B}_Y$.*

Proof. For every $i \in Y$,

$$N_{G[Y]}(i) = N_G(i) \cap Y.$$

Hence $\pi_Y(S_G) = S_{G[Y]}$. The result follows immediately from Lemma 2.2. $\square$

Lemma 6.2 (Twin-trigger equivalence). *For distinct vertices i, j of a graph G , the two seed edges $e_i(N_G(i))$ and $e_j(N_G(j))$ form the lower half of a diamond if and only if i, j are false twins. They form the upper half of a diamond if and only if i, j are true twins.*

Proof. The seed edges form a lower half exactly when their lower endpoints agree:

$$N_G(i) = N_G(j).$$

This equality forces i and j to be nonadjacent, because otherwise $j \in N_G(i)$ but $j \notin N_G(j)$. It also says that their neighborhoods outside $\{i, j\}$ agree. Thus they are false twins. The converse is immediate from the definition of false twins.

The seed edges form an upper half exactly when

$$N_G(i) \cup \{i\} = N_G(j) \cup \{j\}.$$

The membership of i in the left-hand side forces $i \in N_G(j)$, and similarly $j \in N_G(i)$; hence the vertices are adjacent. After removing i and j , the equality gives identical outside neighborhoods, so they are true twins. Conversely, true twins satisfy the displayed equality. $\square$

Lemma 6.3 (A generating neighborhood seed has twins). *If $|X| \geq 2$ and S_G diamond-generates $\mathcal{B}_X$, then G contains a pair of twins.*

Proof. The set S_G has $|X|$ edges, whereas

$$|E(\mathcal{B}_X)| = |X|2^{|X|-1} > |X|.$$

Thus the closure sequence starting from S_G must add an edge. If the first closure round added nothing, then the sequence would already be stable and no later round could add anything. Therefore some diamond has its lower or upper half contained in the original seed set S_G . Its two triggering edges have distinct directions and are the corresponding two neighborhood seeds. By Lemma 6.2, their vertices are twins. $\square$

Proposition 6.4. *If S_G diamond-generates $\mathcal{B}_X$, then G is a cograph.*

Proof. Let H be any induced subgraph of G with at least two vertices. By Lemma 6.1, the neighborhood seed S_H diamond-generates the Boolean lattice on $V(H)$. By Lemma 6.3, the graph H contains a pair of twins. Thus every induced subgraph of G with at least two vertices has twins, and Lemma 4.1 implies that G is a cograph. $\square$

Example 6.5 (The first obstruction). The path P_4 has no pair of twins. By Lemma 6.2, no two edges of S_{P_4} form the lower or upper half of a diamond. Therefore its first closure round adds nothing, so

$$\text{cl}_{\diamond, V(P_4)}(S_{P_4}) = S_{P_4}.$$

In particular, S_{P_4} is not diamond-generating. This is the smallest example obstructing the cograph condition.

7 Proof of the classification and consequences

Proof of Theorem 1.1. Part (i) is Proposition 3.2. For part (ii), Proposition 5.1 proves that cographs generate, and Proposition 6.4 proves the converse.

For part (iii), let S be a diamond-generating set of cardinality $|X|$. By Corollary 3.4, there is a unique graph G on X with $S = S_G$. Part (ii) implies that G is a cograph. Conversely, every cograph G gives a generating set S_G with exactly one edge in each direction and therefore exactly $|X|$ edges. $\square$

Corollary 7.1. *The map $G \mapsto S_G$ is a bijection from the labeled cographs on $[n]$ to the minimum-cardinality diamond-generating subsets of $E(\mathcal{B}_n)$. Consequently, these two classes have the same cardinality.*

Proof. Surjectivity and the fact that every image is a minimum-cardinality generator are contained in Theorem 1.1. Injectivity follows because the open neighborhoods recover every adjacency of G . $\square$

8 Further questions

The classification concerns cardinality, not the number of synchronous closure rounds. The cograph construction suggests several natural refinements.

Question 8.1. For a cograph G , how can the first round at which $\text{cl}_{\diamond, X}(S_G) = E(\mathcal{B}_X)$ be read from a cotree of G ? Which cographs minimize or maximize this closure time among graphs on a fixed number of vertices?

Question 8.2. What are the inclusion-minimal diamond-generating sets that have more than $|X|$ edges?

Question 8.3. Which analogous minimum-generator classifications hold for cover graphs of other finite distributive or modular lattices?

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